

TUTORIALS

1. Solve $\frac{dy}{dx}(1+e^{2x})dy+(1+y^2)e^x dx$, given that when $x=0, y=1$.
2. Solve $(1+x^2)\frac{dy}{dx}+2xy-4x^2=0$ subject to the initial condition $y(0)=0$.
3. Solve $y dx + (1+x^2) \tan^{-1}x \cdot dy = 0$
4. solve the equation $(1+y^2)dx = (\tan^{-1}y - x)dy$
5. Solve $\cos^2 x \frac{dy}{dx} + y = \tan x$.
6. Solve $\frac{dy}{dx} + y = 1$
7. Solve $(1+y^2) + (x - e^{-\tan^{-1}y})\frac{dy}{dx} = 0$.
8. Solve $\frac{dy}{dx} + (2x \tan^{-1}y - x^3)(1+y^2) = 0$.
9. Solve $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$.
10. solve the equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{4x}$
11. solve the equation $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 9y = 40\sin 5x$
12. solve the equation $(D^3 + 1)y = (e^x + 1)^2$
13. solve the equation $(D^2 - 4D + 6)y = x^2 + e^x + \cos 2x$
14. solve the equation $(D^2 - 5D + 6)y = x^2e^x$
15. solve the equation $(D^2 - 2D + 5)y = e^{2x}\sin x$ A
16. solve the equation $(D^2 + 4)y = e^x + \sin x$
17. solve the equation $(D^2 - 2D + 1)y = x\sin x$ B
18. solve the equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 8x^2e^{2x}\sin 2x$
19. solve the equation $x^2\frac{d^2y}{dx^2} - x\frac{dy}{dx} - 3y = x^2\log x$ A
20. solve the equation $x^2\frac{d^2y}{dx^2} + 5x\frac{dy}{dx} + 4y = x\log x$ B
21. solve the equation $\frac{dx}{dt} + 5x + y = e^t$ and $\frac{dy}{dt} - x + 3y = e^{2t}$
22. Solve $\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$ given that $x=2$ and $y=0$, when $t=0$. B

UNIT -II

1. Find the general solution of $(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 3y = 0$ if $y=x$ is a one solution. I A
2. solve $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$ Given that solve $(x + \frac{1}{x})$ is one integral. I
3. solve the general solution of $\frac{d^2y}{dx^2} - \cot x \frac{dy}{dx} - (1 - \cot x)y = e^x \sin x$ I
4. solve the equation $\frac{d^2y}{dx^2} - \frac{2}{x}\frac{dy}{dx} + \left(1 + \frac{2}{x^2}\right)y = xe^x$ I B

5. solve the general solution of $x \frac{d^2 y}{dx^2} - 2(x^2 + x) \frac{dy}{dx} + (x^2 + 2x + 2)y = 0$ IV
6. solve the general solution of $\frac{d^2 y}{dx^2} - 2 \tan x \frac{dy}{dx} + 5y = e^x \sec x$ IV
7. Solve $x \frac{d^2 y}{dx^2} - \frac{dy}{dx} - 4x^3 y = 8x^3 \sin x^2$ IV
8. solve by the variation of parameter $\frac{d^2 y}{dx^2} + 4y = 4 \tan 2x$ IV
9. solve by the variation of parameter. $\frac{d^2 y}{dx^2} + n^2 y = \sec nx$
10. solve by the variation of parameter $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2 e^x$
11. solve by the variation of parameter $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} = e^x \sin x$ IV
12. solve the series of the following equation $(1 + x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0$
13. solve the series of the following equation $(1 - x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + 4y = 0$
14. solve the series of the following equation $2x^2 \frac{d^2 y}{dx^2} + (2x^2 - x) \frac{dy}{dx} + y = 0$
15. solve the series of the following equation $x \frac{d^2 y}{dx^2} + \frac{dy}{dx} - y = 0$

UNIT -III

1. From the partial differential equation by elimination of arbitrary function from the relation $z = y^2 + 2f\left\{\frac{1}{x} + \log y\right\}$
2. From the partial differential equation by elimination of arbitrary constant from the relation $z = (x^2 + a)(y^2 + b)$.
3. solve the differential equation $y^2 zp + x^2 zq = xy^2$
4. solve $px + qy = 3z$
5. solve the differential equation $(z^2 - 2yz - y^2)p + (xy + zx)q = (xy - zx)$
6. solve the differential equation $x^2 p + y^2 q = \frac{xy}{x+y}$ $\frac{xy}{x+y} = \frac{xy}{x+y}$
7. solve the differential equation $p + 3q = 5z + \tan(y - 3x)$
8. solve the differential equation $(y^2 + z^2 - x^2)p - 2xyq = (xy - zx)$ $2zx$
9. solve the differential equation $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$
10. solve the differential equation $(mz - ny)p + (nx - lz)q = (ly - mx)$
11. solve by charpit method $(p^2 + q^2)y = qz$
12. solve by charpit method $z = px + qy + p^2 + q^2$

13. solve by charpit method $q = px + p^2$
14. solve by charpit method $2(z+xp+yq)=yp^2$
15. Solve $\frac{\partial^3 z}{\partial x^3} - 2 \frac{\partial^3 z}{\partial x^2 \partial y} = 3x^2 y$.
16. Solve $\frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x \partial y} + 4 \frac{\partial^2 z}{\partial y^2} = e^{2x+y}$.
17. solve the differential equation $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = \cos x \cos y$
18. solve the differential equation $(r + s - 6t) = y \cos x$
19. Solve $(D^2 - DD' - 6D')z = xy$.
20. Solve $(D^2 + DD' + D')z = \sin(x+2y)$.
21. solve the differential equation $\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = (x - y)$
22. solve the liner partial differential equation $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = e^{x-2y}$
23. solve the linear partial differential equation $2 \frac{\partial^2 z}{\partial x^2} - 5 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = 5 \sin(2x + y)$
24. Solve the linear partial differential equation $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 12xy$

UNIT -IV

1. If $f(z)$ is regular function of z show that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 = 4 |f'(z)|^2$
2. Show that the function $e^x (\cos y + i \sin y)$ is an analytic or holomorphic function find its derivative.
3. Show that the function $u = \frac{1}{2} \log(x^2 + y^2)$ is harmonic function and find its harmonic conjugate.
4. Show that $e^x (x \cos y - y \sin y)$ is a harmonic function find the analytic for which $e^x (x \cos y - y \sin y)$ is imaginary part .
5. Prove that $u = x^2 - y^2 - 2xy - 2x + 3y$ is harmonic. find a function v such that $f(z) = u + iv$ is analytic. Also express $f(z)$ in term of z .
6. Show that the function $u = e^{-2xy} \sin(x^2 - y^2)$ is harmonic find the conjugate function v and express $u + iv$ as an analytic function of z .
7. if $u - v = (x - y)(x^2 + 4xy + y^2)$ and $f(z) = u + iv$ is an analytic function of $z = x + iy$ find $f(z)$ In z of milne's Thomson method.

Cauchy's Integral Formula

8. prove that $\int_c \frac{f(z)dz}{z-a} = 2\pi i f(a)$ When C is the circle, a belongs to C.
9. Evaluate $\int_c \frac{3z^2 + z}{z^2 - 1} dz$ where c is the circle $|z-1| = 1$
10. Evaluate the integral using Cauchy integral formula $\int_c \frac{e^{2z}}{(z-1)(z-2)} dz$ when C is $|z| = 3$
11. Evaluate $\int_c \frac{\cos \pi z^2}{(z-1)(z-2)} dz$ where c is the circle $|z| = 3$
12. Using Cauchy integral formula proved that $\int_c \frac{e^{2z}}{(z+1)^4} dz = \frac{8\pi i}{3e^2}$ where C is the circle $|z| = 3$
13. Find the order of each pole and residue at it of $\frac{1-2z}{z(z-1)(z-2)}$
14. Find the order of each pole and residue at it of $f(z) = \frac{z^2}{(z-1)^2(z+2)}$
15. Determine the pole of the following function and residues at each pole
 $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ And hence evaluated $\int_c \frac{z^2 dz}{(z-1)^2(z+2)}$ where $c: |z| = 3$
16. To Evaluate $\int_0^{2\pi} \frac{d\theta}{5+3\sin\theta}$ by contour integration.
17. To Evaluate $\int_0^{2\pi} \frac{\cos 2\theta d\theta}{5+4\cos\theta}$ by contour integration

UNIT-V

1. proved that $\text{div grad } r^m = m(m+1)r^{m-2}$
2. what is the directional derivative of $\phi = xy^2 + yz^3$ at the point (2,-1,1) in the direction of the normal to the surface $x \log z - y^2 = -4$ at (-1,2,1)
3. Find the directional derivative of $\phi = x^2 - y^2 + 2z^2$ at the point p(2,-1,1) in the direction of the line PQ where Q is the (5,0,2).
4. Find the directional derivative of $\phi = 5x^2y - 5y^2z + \frac{5}{2}z^2x$ at the point p(1,1,1) in the direction of the line $\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z}{1}$.
5. Find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$ when $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$.
6. Show that the vector $\vec{F} = (\sin y + z)i + (x \cos y - z)j + (x - y)k$ is irrotational.
7. proved that $\nabla^2 r^n = n(n+1)r^{n-2}$ where $\vec{r} = xi + yj + zk$

8. proved that $\nabla^2 f(r) = f'(r) + \frac{2}{r} f(r)$
9. Evaluate $\iint_S \vec{F} \cdot \hat{n} ds$ where $\vec{F} = (x + y^2)\hat{i} - 2x\hat{j} + 2yz\hat{k}$ and S is the surface of the plane $2x + y + 2z = 6$ in the first octant.
10. Evaluate $\int_S \vec{F} \cdot d\vec{s}$ where $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ and S is the surface bounding the region $x^2 + y^2 = 4$, $z = 0$, and $z = 3$.
11. Evaluate $\iint_S \vec{F} \cdot d\vec{s}$ over the surface of the region above the xy -plane bounded by the cone $z^2 = x^2 + y^2$ and the plane $z = 4$ if $\vec{F} = 4xzi + xyz^2j + 3zk$.
12. Verify Stokes's theorem for the function $\vec{F} = x^2i + xyj$ integrated along the square whose sides are $x = 0$, $y = 0$, $x = a$ and $y = a$ in the plane $z = 0$.
13. For $\vec{F} = (x^2 - y^2)\hat{i} + 2xy\hat{j}$ taken around the rectangle in the plane $z = 0$ and bounded by the lines $x = 0$, $y = 0$, $x = a$ and $y = a$.
14. For $\vec{F} = (2x^2 - 3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$ then $\iiint_V \nabla \cdot \vec{F} dv$ Evaluate where V is bounded by $x = 0$, $y = 0$, $z = 0$ and $2x + 2y + z = 4$.
15. Verify Divergence theorem for $\vec{F} = x^2i + zj + yzk$ taken over the cube bounded by $x = 0$, $x = 1$, $y = 0$, $y = 1$ and $z = 0$, $z = 1$.